Bias recipe
Mathematics and implementation
Version 1 — review copy

Wenu-imaging

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1 Executive summary and authority

The recipe estimates a master bias *signal* by correcting the electronic scan level separately in every input row, trimming the active rectangle, rejecting entire frames by their corrected active-area means, and stacking the retained frames with the existing Eclipse pixel median. QC describes the inputs, corrections and master; it does not silently modify the result.

Shared input contract. Run `mefpp_recipe cal bias -config RECIPE.toml` from orchestration. The recipe references `project.toml`, numerical settings, an absolute `inputs.list` path and a fresh reduced `run`. Each list row contains an absolute FITS path under the project raw root and a `BIAS` classification. The list directory is configurable. Calibration paths use the project calibration root. Optional BPMs are already reduced products. Dependency-list reduction is deferred. The native archive preserves the classified list. Legacy `--bias` lists remain compatible; new recipes use the same list contract as BPM/linearity.

This guide explains the implemented estimators and their source mapping. Sparse source landmarks identify scientific stages without depending on line numbers. Validation results and work summaries belong in GitHub issues and pull requests; the guide describes the algorithm rather than its development history.

This guide incorporates the already accepted average/lower temporal-median choice and explicit zero-spread refusal. Optional clipped raw-pair RON is explained in the companion guide `mathematics/mefpp/bias-clipped-ron.tex` and its matching PDF; the median supplement is `mathematics/mefpp/bias-median-even-handling.tex` and its matching PDF. The designs in Section 15 retain the scientific authority.

All signal levels, standard deviations and read-noise diagnostics are in ADU. Pixel and frame counts are dimensionless counts. The recipe performs no gain conversion, flat normalization, interpolation or dark-current estimate. It does not provide a master variance or a covariance model.

Three distinctions to retain

1. The detector BPM selects physical active pixels. It is not applied to electronic prescan or overscan samples.

2. Frame rejection selects complete exposures. The subsequent median stack still evaluates every output pixel, including pixels excluded by that BPM.

3. Temporal stacking uses the configured average/lower even-sample policy. Modern checked QC uses an averaged central pair; legacy telemetry helpers retain their documented conventions. These roles must not be conflated.

Mathematical description versus implementation

Equations below state the estimators. Exact reproduction also requires the documented order-statistic convention, evaluation order and pixel/float conversions. Checked statistics retain the existing sum-of-squares implementation; the guide does not substitute a different summation algorithm.

Reading the source

Start with the 60-line coordinator and its three labelled phases in `main`. The preparation, input, processing and publication modules expose the mechanics; `mef_recipe_run` owns common paths, logging and workspace only. Follow one detector through `process_bias_detector`, the file adapter, the in-memory service, and the small stacking kernel. Read configuration and serialization helpers after the scientific flow is clear. All paths below are relative to `mefpp2/ins/mef/`; `common/` contains services and `recipes/` contains the command coordinator.

A landmark is a comment in the source, not an executable C label. Search for its exact bracketed identifier. Each occurs once in source; the guide can reference the same stage for several equations. Comments mark scientific boundaries and explain intent, while ordinary statements remain uncluttered.

Landmark and purpose	Source file and routine
`[BIAS-RECIPE-ADMISSION]` Configuration, input identities and all detector bindings	`recipes/mefpp_bias.c` `main`
`[BIAS-DETECTOR-LOOP]` Independent detector calibration, no mosaic balancing	`recipes/mefpp_bias.c` `main`
`[BIAS-PRODUCT-PUBLICATION]` Complete-detector barrier, final MEF then ASDF	`recipes/mefpp_bias.c` `main`
`[BIAS-DETECTOR-CALIBRATION]` One detector's service call and result ownership	`common/mef_bias_process.c` `process_bias_detector`
`[BIAS-INPUT-LOADING]` Load explicit bound detector HDUs	`common/mef_bias_files.c` `load_bias_frames`

The coordinator owns recipe-wide resources. The file adapter owns loaded raw frames and releases each after its last raw measurement and private copy; the returned result owns its master and QC. Read the following scientific stages before exploring the argument parser or metadata serializers.

Inside one detector

Landmark and purpose	Source file and routine
[BIAS-RAW-PAIR-RON] Noise from borrowed raw frames, before admission	common/mef_cal_bias.c mef_cal_bias_measurements
[BIAS-SCAN-CORRECTION] Private frame copies, row correction and trimming	common/mef_cal_bias.c mef_cal_bias_measurements
[SCAN-ROW-CORRECTION] The shared electronic-row estimator and subtraction	common/mefp_lib.c mef_overscan_correction_traced
[BIAS-FRAME-SELECTION] Corrected frame means, one strict interval, zero-spread stop	common/mef_bias_kernel.c mef_bias_stack_masked_diagnostic
[BIAS-TEMPORAL-STACK] Configured median of retained frames at every active pixel	common/mef_bias_kernel.c mef_bias_stack_masked_diagnostic
[BIAS-MASTER-QC] Master statistics and diagnostic high-offset flags	common/mef_cal_bias.c mef_cal_bias_measurements
[BIAS-DETECTOR-WRITE] Staged detector MEF and exclusive publication	common/mef_bias_files.c publish_bias_header

Supporting routines. common/mef_bias_config.c, mef_bias_expand, handles the TOML envelope and detector selection. common/mef_bias_product_qc.c, mef_bias_product_qc_add, copies measured values, masks and provenance into the common archive. common/mef_bias_qc_fits.c, mef_bias_qc_fits_header, serializes that stored QC into additive FITS cards. These supporting routines do not choose new pixels or rerun the scientific estimators.

The first reading pass retains existing operation/ownership boundaries. Long validation and coordinator routines are separated into logical paragraphs; ordinary argument handling uses explicit control flow. Further extraction should be driven by the owner's source review rather than an arbitrary function-size limit. Passing numerical checks alone does not complete that review.

2 Identity, coordinates and admitted inputs

For one logical detector let $f = 0, \ldots, F-1$ identify supplied raw frames in their original order, and let $D_f(x, y)$ be a loaded value in ADU. FITS scaling belongs to the existing loader; values entering this service are the resulting `pixelvalue` samples.

Logical detector ID, source HDU, zero-based configuration index and output HDU are independent bindings. In the retained OmegaCAM run CCD65 used configuration index 0 and output HDU 1; CCD69 used index 16 and output HDU 17. The actual source HDU is checked in each input, not inferred from its ordinal.

The trim rectangle is inclusive and one-based:

$$\mathcal{R} = [x_0, x_1] \times [y_0, y_1], \qquad W_A = x_1 - x_0 + 1, \quad H_A = y_1 - y_0 + 1.$$

An active pixel (i, j) refers to raw coordinate

$$(x, y) = (x_0 + i - 1,\ y_0 + j - 1). \tag{1}$$

The code converts coordinates at the array-access boundary. This pixel identity does not validate the legacy celestial WCS or supply the deferred detector-layout fallback.

A binary BPM has an explicit HDU, polarity, coordinate domain, readout and binning. The loader normalizes the declared good value to Eclipse polarity $M = 1$ for good and $M = 0$ for excluded. Active-sized masks are embedded using Equation (1); raw-sized masks are cropped consistently for active measurements. Explicit omission is recorded; no mask is guessed from a filename.

Validation before calculation

The service checks operation/version, detector/request bindings, common image dimensions, scan/trim bounds, rejection counts, finite nonnegative frame parameter κ, positive diagnostic parameter k, and minimum input count. The configured minimum is at least three. The entire loaded raw plane must contain finite values, including masked pixels. Malformed masks or contradictory calibration declarations fail.

General statistics helpers can report nonfinite exclusions; this capability does not relax the finite-raw-input rule. No raw saturation threshold or automatic exposure discovery is added by this bias operation.

Code boundary

`mef_layout_load`, `mef_layout_load_bpm` and binding checks handle file/geometry identity. `mef_cal_bias_measurements` validates and calculates one detector using copies, leaving caller raw planes and geometry unchanged.

3 Electronic row correction

For a raw row y, concatenate the enabled prescan and overscan column bands into a vector s_y of length n_s. No detector BPM is applied. If both bands overlap, their common samples enter twice; the configuration must describe the intended electronic samples. Disabled bands contribute nothing.

Sort the samples as $s_{y,(1)} \le \cdots \le s_{y,(n_s)}$. With integer low and high rank-rejection counts ℓ, h, the row correction is

$$c_f(y) = \mathrm{pix}\left\{ \frac{1}{n_s - \ell - h} \sum_{r=\ell+1}^{n_s-h} s_{y,(r)} \right\}, \qquad n_s - \ell - h > 0. \tag{2}$$

The sum uses double precision; the result is cast to the pixel type. These are rank cuts, not iterative sigma clipping. Every column is corrected; then the active rectangle is extracted:

$$A_f(i,j) = \mathrm{pix}\{D_f(x_0 + i - 1, y_0 + j - 1) - c_f(y_0 + j - 1)\}. \tag{3}$$

The retained OmegaCAM configuration enables overscan only. CCD65 columns 2100–2142 contain 43 samples; rejecting ten at each end leaves 23. Its active rectangle is $[51, 2094] \times [5, 4095]$, or 2044×4091 pixels.

The correction profile covers every raw row, including rows outside the trim. Its legacy level is the pixel-median (Section 5); its reported RMS is a sample standard deviation about its own mean:

$$s_c = \sqrt{\frac{\sum_y (c_f(y) - \bar{c}_f)^2}{H_{\mathrm{raw}} - 1}}.$$

The implementation subtracts the median before the centered sum-of-squares evaluation. This shift preserves the exact estimator. Legacy summaries pass through float fields. Median selection reorders the working profile; scatter is invariant to reordering. The bias record stores summaries, not an ordered profile array. Separate raw scan QC uses enabled columns over trimmed rows.

`mef_overscan_correction_traced` corrects and crops; `function1d_average_reject_checked` implements Equation (2).

4 Frame admission and temporal stack

Source: `mef_bias_kernel.c`, `mef_bias_stack_masked_diagnostic`, `[BIAS-FRAME-SELECTION]` and `[BIAS-TEMPORAL-STACK]`.

For each corrected active image, use all active pixels without a BPM, or only its good pixels otherwise. The per-frame mean is

$$a_f = \frac{1}{N_f} \sum_{(i,j)\in\mathcal{G}_f} A_f(i,j).$$

An empty valid population fails. The rejection band uses all F submitted frame means once:

$$\bar{a} = \frac{1}{F}\sum_f a_f, \qquad s_a = \sqrt{\frac{\sum_f (a_f - \bar{a})^2}{F-1}}, \tag{4}$$

$$L = \bar{a} - \kappa s_a, \qquad U = \bar{a} + \kappa s_a. \tag{5}$$

The frame-band implementation accumulates centered squared deviations in a second pass. Retain only $L < a_f < U$; equality at either boundary is rejected. The band is not recalculated after rejection. The configured minimum number of retained frames must be met. By the owner decision of 8 November 2026 (issue #289), exactly $s_a = 0$ is an explicit computation failure before admission or stacking: the collapsed interval cannot admit any frame. The native and legacy bias callers report the detector ID, every input filename and its computed corrected mean (17 significant digits, ADU). The native service also returns a specific zero-spread error and transfers no master. The recipe stops before final FITS/ASDF publication and retains failure evidence. The test uses the computed double-precision s_a, never its rounded legacy float QC representation. No near-zero tolerance or inclusive endpoint rule is introduced; every nonzero-spread case keeps the existing arithmetic. The retained OmegaCAM setting is $\kappa = 2$, minimum three frames.

Historical configuration entries `FRAMETHR.LO`/`FRAMETHR.HI` do not add absolute cuts in this kernel. The computed band is operational. For retained set $\mathcal{K}$, $K = |\mathcal{K}|$, the master is

$$B(i,j) = \mathrm{med}_{\mathrm{configured}}\{A_f(i,j) : f \in \mathcal{K}\}. \tag{6}$$

There is no per-pixel clipped mean or per-pixel rejection count. The BPM selects frame-mean and QC populations; it does not replace output pixels. The temporal median is evaluated at every active coordinate. `mef_bias_stack_masked_diagnostic` selects frames and consumes the private cube; `cube_avg_median_opts` implements the configured stack. The stack allocates an independent master image. Once stacking succeeds, the kernel releases the consumed corrected cube before master-statistics working arrays are allocated, and clears its cleanup pointer. This changes no estimator or numerical evaluation order; failures still release each owner once. Ordinary in-memory calls keep borrowed raw images unchanged through the service call. The file adapter uses `mef_cal_bias_owned_measurements`: all validation and requested raw-pair diagnostics precede consumption; each frame is copied and its raw/scan QC measured before its raw owner is released and nulled, then correction proceeds on the private copy. No later stage reads that raw view. On failure, adapter cleanup destroys remaining non-null owners. Mismatched or shared pixel owners are rejected before consumption. This reduces raw/corrected overlap without changing scientific evaluation order. Releasing the corrected cube removes its overlap with master statistics but cannot remove an earlier full-QC peak during per-frame measurements. Resident memory also includes allocator and mapped-file state; actual-host profiling remains necessary. The synthetic `bias_memory_probe` checks release order in both QC profiles and can capture compiler-local numerical snapshots for ten WFI-sized raw frames; its optional final `owned` argument checks that no raw owners survive to stacking. Failure injection covers partial consumption and caller cleanup. QC retains original input IDs.

For synthetic frame means $0, 1, 2, 10$ with illustrative $\kappa = 1$, $\bar{a} = 3.25$, $s_a = \sqrt{62.75/3} \simeq 4.573474$. The band is approximately $[-1.323474, 7.823474]$; the first three survive. This example uses $\kappa = 1$ to expose the rule, not the retained setting 2.

5 Median conventions and precision

For sorted $z_{(1)} \le \cdots \le z_{(N)}$, the modern checked QC median is

$$m(z) = \begin{cases} z_{((N+1)/2)}, & N \text{ odd}, \\ \frac{1}{2} z_{(N/2)} + \frac{1}{2} z_{(N/2+1)}, & N \text{ even}. \end{cases}$$

The bias temporal stack defaults to `median_even_handling="average"`. For even K, it uses $\frac{1}{2} z_{(K/2)} + \frac{1}{2} z_{(K/2+1)}$, evaluated in double precision and converted once to `pixelvalue`. Explicit `"lower"` uses $z_{(K/2)}$ for historical bias compatibility. Both modes use the central order statistic for odd K. The bias stack requires at least three retained frames. Selection follows the Wirth implementation documented in the source [1]; the existing timeline buffer is reused.

Ten retained frames therefore use the midpoint of the fifth and sixth sorted samples by default, or the fifth in lower mode; nine frames use the fifth in both modes. Generic Eclipse median callers retain their historical behavior: the old pixel helper returns the lower central value for even $K \ge 4$, while its two-value case averages. The historical `double_median` helper uses the upper middle for even samples. These helpers must not be substituted for the configured temporal estimator or the modern checked QC median.

Legacy master statistics

With a BPM, the kernel overrides the historical master median and standard deviation with checked *masked* statistics, then converts results to float. They are not inherently unmasked. Without a BPM, the historical median uses the pixel-median convention; modern QC averages the middle pair. The explicitly named `master_unmasked` family always omits the BPM. Even for equal definitions and populations, modern double results and legacy float fields can differ through rounding.

The default `pixelvalue` is float; `DOUBLEPIX` selects a different build lane. Scan sums use double precision but are converted to pixel type before subtraction. Default FITS pixels are IEEE single precision. Existing legacy cards retain six-decimal formatting; new scalar cards use 17 significant digits with a FITS-compatible uppercase exponent. A short printed table cannot establish exact numerical equality.

Worked order-statistic examples

For $0, 2, 4, 6$, modern QC and default temporal median are 3; explicit temporal lower mode gives 2. For scan samples $100, 101, 102, 103, 200$ and one rank rejected at each end, the correction is 102. Raw samples 110 and 112 become 8 and 10.

6 QC calculation profiles

The standalone recipe accepts `-qc full` or `-qc legacy`. The TOML equivalent is `[bias] qc = "full"` or `qc = "legacy"`. An explicit command-line choice overrides the TOML choice; omission preserves `full`. Unknown or duplicate command-line choices fail before processing.

Full QC defaults to MAD off: omitted `[bias] mad` means `mad = false`, and omitted long-form `--mad` means `no`. Explicit `mad = true` or long-form `--madyes` opts in. Clipped RON remains independent of this choice and follows its existing configuration. MAD-based scatter, MAD RON and the MAD-based hot-pixel diagnostic are unavailable when MAD is off; hot-pixel thresholds have not been replaced by sigma-clipping. Other measurements keep their definitions. Legacy QC retains the row-correction median/scatter and scan parameters, frame-admission summaries and master median/scatter already produced by the reduction. It skips the added raw-active, corrected-active and scan-region sample statistics, raw-pair RON, clipped RON, MAD and high-offset diagnostics. Expanded records remain present with null values and the reason `disabled_by_legacy_qc`; they are not measured zeros. Cheap configuration provenance remains available. The profile takes precedence over MAD/RON measurement requests.

This is a measurement choice, not a pixel-processing preset. Input validation, configured detector BPM use in frame admission, correction geometry, rejection rules, temporal median convention and the existing master summary estimators remain unchanged. In particular, requesting legacy QC does not silently replace current masked frame means with the historical unmasked means or select the lower temporal median. Comparisons with a historical executable still require explicit matching settings and pixel checks.

Legacy products record `provenance.qc_profile = "legacy"` in ASDF and `ESO MPP QC PROFILE` in FITS. Absence of this optional field means full QC for compatibility with existing products. Both modes retain FITS, ASDF and log publication, membership/provenance and mandatory validation; the switch allows numerical-QC comparisons within the same processing and publication pipeline. Skipped diagnostics also reduce their associated serialization work.

7 Checked sample statistics and MAD

A statistic defines its rectangle, processing stage, mask and nonfinite policy. Within that rectangle, BPM-excluded pixels are counted first; among remaining pixels, nonfinite samples are excluded. Thus region count equals BPM-excluded plus nonfinite-excluded plus valid count. For valid values $z_1, \ldots, z_N$,

$$\bar{z} = \frac{1}{N}\sum_i z_i, \qquad s_z = \sqrt{\frac{\sum_i (z_i - \bar{z})^2}{N-1}}, \tag{7}$$

$$d_z = m(\{|z_i - m(z)|\}), \qquad \mathrm{sMAD}(z) = 1.4826\, d_z. \tag{8}$$

Both medians in d_z use the averaged-middle convention. The raw MAD is scaled exactly once. The constant approximates $1/\Phi^{-1}(3/4)$, making scaled MAD consistent for normal-distribution standard deviation at large sample size [2, 3]. It is not a finite-sample unbiasedness correction, nor does its use establish Gaussian detector noise.

Unlike the two-pass frame-band scatter, per-pixel sample helpers evaluate

$$v = \frac{\sum_i z_i^2 - (\sum_i z_i)^2/N}{N-1}, \qquad s_z = \sqrt{\max(0, v)}.$$

They accumulate in double precision. Cancellation can affect nearly constant, large-offset data; this guide records the current arithmetic rather than replacing it with Welford accumulation. Legacy scan scatter uses the same centered sum-of-squares family after its median shift.

Situation	Representation
$N = 0$	Level and scatter unavailable; not measured zero.
$N = 1$	Mean/median measured; requested MAD is zero; sample stdev unavailable.
MAD omitted	Unavailable by default or explicit choice; not zero.
MAD equals zero	Valid measured result; may make a derived threshold degenerate.

The historical `image_median_stat` computes a mean absolute deviation; it must not be substituted for this median absolute deviation. `image_getstats_masked_checked` selects the population; `double_median_mad_checked` calculates checked median and MAD.

The selective entry point `image_getstats_masked_selected_checked` accepts independent `IMAGE_SAMPLE_MEAN`, `IMAGE_SAMPLE_STDEV`, `IMAGE_SAMPLE_MEDIAN` and `IMAGE_SAMPLE_MAD` request bits. Population counts and mask/zone metadata are always returned. Unrequested values remain zero; availability applies only to requested results. `level_available` covers a requested mean or median, with the request identifying which value is present. MAD needs an internal median even when the caller does not request that median as an output. Empty/unknown requests fail without changing the output record.

The original checked entry point retains mean, sample scatter, median and optional MAD. Both paths retain the existing population and accumulation order. The internal legacy statistics pass omits its discarded median; checked median/MAD still use the same double-precision helper. Bias frame admission requests only the masked mean, avoiding unused scatter and order-statistic work. An empty population still fails frame admission. No stored QC quantity or estimator is removed.

For $0, 2, 4, 6$, $s = \sqrt{20/3} \simeq 2.581989$, MAD $= 2$ and scaled MAD $= 2.9652$.

8 Which pixels belong to each diagnostic?

Family	Samples and stage	Mask
`raw_active`	Raw input, active trim rectangle	Declared detector BPM, or explicit omission
`corrected_active`	Row-corrected, trimmed input	Cropped active BPM, or omission
Prescan / overscan	Raw enabled scan band, trimmed rows	Detector BPM not applicable
Scan correction	Row corrections, full raw height	Detector BPM not applicable
`master`	Final stacked active image	Active BPM, or omission
`master_unmasked`	Same stacked active image	Explicitly no detector BPM
`raw_difference`	Original raw-pair difference, active rectangle	Intersection of both good masks

No clipping is silently added to these populations. Electronic rank rejection belongs to row correction, and whole-frame rejection belongs to stacking. Their QC context identifies these distinct stages.

Per-input scan and active measurements are retained for all submitted frames, including rejected ones. Consequently a range over all ten inputs need not summarize only the nine used for a particular detector.

Master spatial scatter describes differences across detector pixels, including structure and residual offsets. It is not the uncertainty of each master pixel. The difference between masked and unmasked scatter can be scientifically useful; neither is inherently a substitute for the other.

9 Read noise from the explicit raw pair

Two distinct, explicitly selected original raw frames a, b supply

$$\Delta(x,y) = D_a(x,y) - D_b(x,y), \qquad \mathrm{RON}_s = s_\Delta/\sqrt{2}, \quad \mathrm{RON}_{\mathrm{MAD}} = 1.4826\, d_\Delta/\sqrt{2}.$$

The population is the active rectangle with the intersection of good masks. This is measured *before* row correction or master subtraction. Selecting a raw pair is independent of later stack admission: either selected frame can be rejected from the master. `image_pair_stats_checked` forms the pixel-type difference and invokes checked sample statistics.

For independent equal-variance noise, differencing doubles the variance. More generally,

$$\mathrm{Var}(D_a - D_b) = \mathrm{Var}(D_a) + \mathrm{Var}(D_b) - 2\,\mathrm{Cov}(D_a, D_b).$$

The factor $\sqrt{2}$ therefore has an interpretation conditional on independence and equal variance. Stable spatial bias structure cancels; temporal row drift or other changing structure can contribute. Centering the difference for a standard deviation is not equivalent to electronic row correction. The generic sampled-window `image_rect_readout_noise` routine is a different estimator and is not used for this bias diagnostic.

Quantization and repeated robust values

For raw differences $-2, 0, 2, 4$, the mean is 1 and

$$s_\Delta = \sqrt{20/3}, \quad \mathrm{RON}_s = \sqrt{10/3} \simeq 1.825742, \quad d_\Delta = 2, \quad \mathrm{RON}_{\mathrm{MAD}} \simeq 2.096713.$$

Repeated values near 2.0967 ADU across detectors can result from the same integer-valued MAD of 2. They do not demonstrate identical underlying noise.

Master uncertainty is not supplied

This recipe supplies no master variance or covariance plane. In particular, RON^2/K is not its implemented uncertainty model: the temporal stack is a median, and row correction introduces shared estimates. For pixels p, q corrected using the same estimated row offset c,

$$\mathrm{Cov}(D_p - c, D_q - c) = \mathrm{Cov}(D_p, D_q) - \mathrm{Cov}(D_p, c) - \mathrm{Cov}(c, D_q) + \mathrm{Var}(c).$$

This identity explains a possible correlation; the current product does not estimate its terms or assume their independence.

10 High-offset pixels

With requested MAD, the selected master population gives median m_B and scaled MAD $s_{B,\mathrm{MAD}}$. For configured positive k,

$$T = m_B + k\, s_{B,\mathrm{MAD}}, \qquad H(i,j) = \mathbf{1}\{B(i,j) > T\}, \qquad N_H = \sum_{i,j} H(i,j).$$

The threshold population uses the declared BPM. The subsequent search visits *all* output pixels, including BPM-excluded ones, and uses strict $>$. The direct flag loop has no additional finite-value filter. Finite raw-input validation does not prove that every hypothetical floating-point overflow in later arithmetic would remain finite.

Flag polarity is 1 = flagged, unlike the normalized good-pixel BPM polarity. Coordinates are one-based in the active output. This diagnostic does not create or apply a new detector BPM. A high bias offset is not a dark-current measurement, and $k = 5$ does not by itself define a calibrated false-alarm rate.

A zero MAD is a measured but degenerate scale: the threshold becomes the median and every larger output value qualifies. With MAD omitted the derived threshold/count are unavailable. A measured zero count remains zero; an empty coordinate set is represented with its explicit availability/reason context.

For master values $0, 1, 1, 2, 20$ with no exclusions, median and MAD are both 1. At $k = 5$, $T = 1 + 5(1.4826) = 8.413$ and only 20 is flagged. If 20 is BPM-excluded, the threshold is recomputed from the remaining population, but the final search still considers 20. This distinction is deliberate.

11 FITS and ASDF: one measurement, two representations

The accepted policy retains existing FITS QC cards and writes new scalar diagnostics to both formats. Measurements and context are prepared from the same common QC record. Neither writer independently remeasures pixels. `mef_bias_qc_fits.def` is the authoritative field-to-card mapping. The following summarizes families, rather than replacing that mapping.

FITS family / field	Meaning in common QC
Legacy `FRMEDIAN`, `FRSTDEV`	Historical master statistics; retain their precision and semantics
`MAST`, `UMAS`, `DIFF`	Selected master, unmasked master, raw difference; mean, median, stdev and scaled MAD
`RON`	Raw-pair standard-deviation and MAD estimators
`HOT`	Count, threshold, multiplier and zero-scale/degeneracy context
Per-frame `RAW`, `CORR`	Raw and corrected active statistics
Per-frame `PRES`, `OVER`	Enabled electronic scan populations
Legacy scan fields	Correction-profile median and sample scatter

A statistic carries unit, status/reason, stage, region, coordinate system, mask identity/applicability, clipping and nonfinite policies, sample count and source record references as applicable. An unavailable measurement has no invented numeric card. Measured zero and unavailable are distinct.

The verified real product contained 89,445 mapped scalar/context cards: $32 \times 2795 + 5$. These are not 89,445 independent numerical diagnostics. Five detectors had one additional rejection-reason card. Arrays, embedded input records and complete provenance remain in ASDF. FITS header scalars alone do not reconstruct a full provenance tree. A future harvester must preserve units, state and definitions, not merely collect numeric keyword values.

Version domains

The retained native library version 0.3.0, ASDF file/core 1.6, application schema 2.0.0 and mapping identifier `bias-scalar/1` refer to different version domains. They must not be compared as though they were one software version. The native interface includes the qualified memory-backed readback path for the relevant library limitation.

12 Operation, publication and explicit recovery

Normal operation keeps the raw directory untouched, launches orchestration from its own directory and writes final reduced products to the explicit product directory. In the retained run these were respectively `/Users/fselman/Pictures/OCAM/bpm`, `/Users/fselman/Pictures/OCAM/test_recipes/orchestration`, and `/Users/fselman/Pictures/OCAM/test_recipes/reduced/cal/bias`.

The standalone coordinator processes detectors serially: load and validate, measure the raw pair, correct private frame copies, compute input QC, reject frames, stack, and prepare common QC/header data. It writes private detector products and releases detector-sized arrays. A single native writer assembles the final MEF; the native QC writer publishes ASDF. A successful recipe requires the required outputs to complete. Exclusive publication prevents silent replacement; the FITS/ASDF/log set is not an atomic three-file transaction.

Expanded scalar/context headers exposed a former fixed 1 MiB aggregate readback budget. The fix budgets the exact emitted, padded header bytes with checked arithmetic while retaining the individual source-header bound. Image payload bytes are not charged to the header budget. This repairs an assembly limit; it does not change measured pixels or QC estimators.

The retained failed run was explicitly recovered from its private detector products after verification. Recovery compared all pixel payloads and legacy QC, checked mapped scalar/context cards against the original QC, then wrote a native ASDF staging file and independently validated the complete record, arrays and provenance before exclusive publication. It retained the failed log and recovery identity. It did not retroactively declare the original recipe successful or authorize unrestricted provenance harvesting.

The optional-cleanup change guards caller-side releases of absent optional objects, covers repeated/partial cleanup and creation-name replacement, and preserves allocator diagnostics for actual NULL misuse. It changes neither these equations nor the global allocation contract.

13 Implementation crosswalk

Paths below are relative to `mefpp2/`. The reviewed revision identifies actual source, while names identify the relevant calculation boundary.

Path	Responsibility
`ins/mef/common/mef_cal_bias.c`	Detector measurement service, raw RON, mask populations, hot pixels
`ins/mef/common/mef_bias_kernel.c`	Frame means, one-pass admission policy, median stack, legacy statistics
`ins/mef/common/mefp_lib.c`	Row correction, legacy scan summaries, trim and optional cleanup
`src/math/function_1d.c`	Checked rank-rejected electronic mean
`src/math/median.c`	Pixel median and order-statistic selection
`src/math/dstats.c`	Checked double median/MAD and sample scatter
`src/iproc/cube2image.c`	Temporal per-pixel median
`src/iproc/image_stats.c`	Checked masked/pair populations and statistics
`ins/mef/common/mef_bias_product_qc.c`	Shared measurement/context record construction
`ins/mef/common/mef_bias_qc_fits.c` and `mef_bias_qc_fits.def`	Scalar/context FITS serialization and authoritative mapping
`ins/mef/common/mef_bias_files.c`	File adapter and detector-product preparation
`ins/mef/common/mef_operation_template.c`	Operation lifetime and cleanup
`ins/mef/common/mef_mef_assemble.c`	Native MEF assembly and header budget
`ins/mef/common/mef_product_qc_asdf.c`	Native ASDF writing, readback and explicit recovery
`ins/mef/recipes/mefpp_bias.c`	Standalone serial coordinator

14 Validation

Acceptance compares representative real-data products by recipe: pixels, detector mapping, FITS metadata and saved ASDF QC/provenance. See `docs/acceptance.md` for the maintained policy. Target results and known limitations belong in GitHub issues; successful numerical comparisons alone do not establish detector health or ESO Phase-3 compliance.

15 Approved design basis

Repository paths identify the human-reviewed design lineage. Later decisions supersede earlier scope or serialization arrangements where stated.

1. `docs/architecture/proposed/m1/bias-service.md`: numerical compatibility, operation boundary and ownership.
2. `docs/developer/design/bias-clipping-project-bpm-v1.md`: raw-pair RON populations and explicit BPM policy.
3. `docs/developer/design/bias-mef-native-asdf-v1.md`: final native MEF/ASDF product and raw/orchestration/reduced-directory operation.
4. `docs/developer/design/bias-qc-fits-asdf-mapping-v1.md`: retain legacy cards and serialize new scalar diagnostics to both formats.

This is the post-implementation mathematical account for owner review, not an amendment to the accepted numerical policy. Source discrepancies discovered in future review should be resolved explicitly rather than hidden by changing an equation or a test expectation.

References

[1] N. Wirth, *Algorithms + Data Structures = Programs*, Prentice-Hall, 1976. Algorithm attribution as documented by the retained Eclipse median source; exact even-sample conventions are stated in this guide.

[2] R Core Team, *Median Absolute Deviation*, R stats reference manual, accessed 6 November 2026. https://www.stat.math.ethz.ch/R-manual/R-patched/RHOME/library/stats/html/mad.html. Normal-consistency scaling and median conventions.

[3] NIST, *Median Absolute Deviation*, Dataplot reference manual, accessed 6 November 2026. https://www.itl.nist.gov/div898/software/dataplot/refman2/auxillar/mad.htm. Definition of MAD and normal-scale interpretation.